AMRITA VISHWA VIDYAPEETHAM MYSURU CAMPUS

Amrita School of Computing, Mysuru Campus

COURSE PLAN

School	School of Computing
Department	Computer Science
Course Title	**SQLite**
Course Code	**24CSA787**
Programme and Section	MCA ‘D’
Academic Year	2026-2027
Semester	III
LTP/ hours per week- Credit	**0-0-1-1**
Course Instructor Details	Ms. Kanchana V v_kanchana@my.amrita.edu

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Programme Outcomes:

PO1	Computational Knowledge: Apply knowledge of computing fundamentals, computing specialization, mathematics, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements.
PO2	Problem Analysis: Identity, formulate, research literature, and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.
PO3	Design /Development of Solutions: Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.
PO4	Conduct Investigations of Complex Computing Problems: Use research-based knowledge and research methods, including design of experiments, analysis, and interpretation of data, and synthesis of the information to provide valid conclusions
PO5	Modern Tool Usage: Create, select, adapt and apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations.
PO6	Professional Ethics: Understand and commit to professional ethics and cyber regulations, responsibilities, and norms of professional computing practice.
PO7	Life-long Learning: Recognize the need, and have the ability, to engage in independent learning for continual development as a computing professional.
PO8	Project management and finance: Demonstrate knowledge and understanding of the computing and management principles and apply these to one's work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO9	Communication Efficacy: Communicate effectively with the computing community, and with society at large, about complex computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions.
PO10	Societal and Environmental Concern: Understand and assess societal, environmental, health, safety, legal, and cultural issues within local and global contexts and the consequential responsibilities relevant to professional computing practice.
PO11	Individual and Teamwork: Function effectively as an individual and as a member or leader in diverse teams and in multidisciplinary environments.
PO12	Innovation and Entrepreneurship: Identify a timely opportunity and using innovation to pursue that opportunity to create value and wealth for the betterment of the individual and society at large.

Course Outcomes (COs):

COs	Description
CO1	Examine the usage, features, installation of SQLite
CO2	Implement operations in database, tables
CO3	Use complex queries like joins, Triggers and keys

Lesson Plan

Week	Topics	Keywords	Textbook with Chapter	CO Mapping	Practice Questions	Assignment Questions
1	Introduction to SQLite, Features, Installation and SQL Syntax	SQLite, DBMS, Installation, SQLite CLI, SQLite Browser, SQL Commands	*T1* - Chapter 1: Introduction to SQLite	CO1	1. Install SQLite and verify the installation. 2. Create a database using SQLite command line called school_db.sqlite. 3. Execute basic SQL commands in SQLite. 4. Use CLI to open an existing SQLite database and check the list of tables.	1. Explain the features of SQLite. 2. Compare SQLite with MySQL and other DBMS. 3. Describe SQLite architecture. 4. Explain advantages of using SQLite. 5. List limitations of SQLite.

2	Data Types, Operators and Expressions	INTEGER, TEXT, REAL, BLOB, NULL, Arithmetic Operators, Logical Operators, Expressions	*T1* - Chapter 2: Data Types and Operators	CO1	1. Create a table using different SQLite data types. 2. Execute arithmetic operations using SQL expressions. 3. Implement comparison and logical operators. 4. Validate data entry with different datatypes	1. Explain SQLite storage classes. 2. Demonstrate arithmetic operators with examples. 3. Write SQL queries using logical operators. 4. Differentiate NULL and default values. 5. Develop queries using expressions.
3	Database Operations and Table Management (Create, Attach, Detach, Drop)	CREATE DATABASE, ATTACH DATABASE, DETACH DATABASE, CREATE TABLE, DROP TABLE	*T1* - Chapter 3: Database and Table Management	CO2	1. Create a SQLite database. 2. Attach and detach a database. 3. Create and drop tables.	1. Design a Student database schema. 2. Explain ATTACH and DETACH commands. 3. Create tables for Library Management System. 4. Demonstrate table creation with constraints. 5. Explain DROP TABLE operation.
4	CRUD Operations – Insert, Update and Delete	INSERT, UPDATE, DELETE, Records, Data Manipulation	*T1* - Chapter 4: Data Manipulation Language	CO2	1. Insert records into a table. 2. Update records using conditions. 3. Delete selected records.	1. Insert multiple records into Employee table. 2. Update employee information based on conditions. 3. Delete records using WHERE clause.

						4. Explain CRUD operations with examples.
						5. Develop SQL statements for database modification.
5	Data Retrieval and Conditional Queries	SELECT, WHERE, DISTINCT, LIKE, IN, BETWEEN, ORDER BY	*T1* - Chapter 5: Querying Data	**CO2**	1. Retrieve records using SELECT statement.	1. Write queries using LIKE operator.
					2. Apply WHERE condition.	2. Retrieve records using IN and BETWEEN.
					3. Sort records using ORDER BY.	3. Use DISTINCT to remove duplicate values.
						4. Create queries using multiple conditions.
						5. Generate reports using SELECT queries.
6	Aggregate Functions and Grouping	COUNT, SUM, AVG, MIN, MAX, GROUP BY, HAVING	*T1* - Chapter 6: Aggregate Functions	**CO2**	1. Find record count using COUNT().	1. Calculate department-wise employee salary.
					2. Calculate average values using AVG().	2. Find maximum and minimum values.
					3. Group records using GROUP BY.	3. Explain GROUP BY and HAVING clauses.
						4. Write SQL queries using aggregate functions.
						5. Generate summary reports from database tables.
7	**Lab Test – I**					

8	Joins – Inner, Outer and Advanced Joins	INNER JOIN, LEFT JOIN, RIGHT JOIN, CROSS JOIN, SELF JOIN	*T1* - Chapter 7: Joins	CO3	1. Perform INNER JOIN between two tables.	1. Create related tables and apply joins.
					2. Execute LEFT JOIN query.	2. Compare different types of joins.
					3. Implement SELF JOIN.	3. Retrieve data from multiple tables.
						4. Solve real-world problems using joins.
						5. Generate reports using join queries.
9	Date and Time Functions	DATE(), TIME(), DATETIME(), STRFTIME(), Date Calculation	*T1* - Chapter 8: Date and Time Functions	CO3	1. Display current date and time.	1. Find employees joined within a period.
					2. Format date values.	2. Calculate difference between two dates.
					3. Perform date calculations.	3. Explain SQLite date formats.
						4. Write queries using date functions.
						5. Develop date-based database applications.
10	Keys and Constraints	PRIMARY KEY, FOREIGN KEY, UNIQUE, CHECK, NOT NULL	*T1* - Chapter 9: Constraints	CO3	1. Create a table with primary key.	1. Design tables using appropriate keys.
					2. Implement foreign key constraint.	2. Explain different types of constraints.
						3. Implement referential integrity.

					3. Apply UNIQUE and CHECK constraints.	4. Create tables using composite keys. 5. Develop normalized database design.
11	Triggers	CREATE TRIGGER, BEFORE Trigger, AFTER Trigger, INSERT, UPDATE, DELETE	*T1* - Chapter 10: Triggers	**CO3**	1. Create an INSERT trigger. 2. Create an UPDATE trigger. 3. Test trigger execution.	1. Create audit trigger for database changes. 2. Explain BEFORE and AFTER triggers. 3. Develop validation triggers. 4. Create DELETE triggers. 5. Implement triggers in real-time applications.
12	Integrated SQL Practice – CRUD, Queries, Joins and Constraints	SQL Queries, Database Design, CRUD, Joins, Constraints	Chapters 4–10	**CO2, CO3**	1. Design a complete database. 2. Perform CRUD operations. 3. Execute join queries.	1. Create Hospital Management database. 2. Create Banking Management database. 3. Apply keys and constraints. 4. Generate reports using joins. 5. Implement triggers for automation.

13	Lab Test – II					
14	Mini Project Demonstration, Viva Voce and Record Evaluation	Project Presentation, Database Implementation, SQL Queries	All Chapters	CO1, CO2, CO3	1. Demonstrate database design. 2. Explain SQL queries used in project. 3. Present project outputs.	1. Submit complete database application. 2. Explain database design decisions. 3. Demonstrate CRUD operations. 4. Explain joins, keys and triggers used. 5. Prepare project report and presentation.

TextBook

1 The Definitive Guide to SQLite Grant Allen and Mike Owens (Second Edition)

CO-PO Mapping

PO/PSO CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1			1						2	
CO2	3	2	2									
CO3	2	2			2						2	

Internal Assessment Component: 70 Marks

Sl. No.	Component Name	Weightage	CO Mapping	Tentative Date of Completion
1	Lab Test 1(Test)	20	CO1, CO2	7th Week
2	Lab Test2(Test)	20	CO3	13th Week
3	Observation Book	5	CO1, CO2, CO3	15th Week
4	Certification	5	CO1, CO2, CO3	14th Week
5	Project + Documentation	20	CO1, CO2, CO3	15th Week

External End Sem Exam: 30 Marks

Names and Signatures:

1. **Course Teacher:** Ms. Kanchana V 22/8/2025
2. **Class Representatives:** SANJAY.S (MY.AC P2MCA 25240)
3. **Class Advisor:**
4. **Program Coordinator:**
5. **Chairperson:**